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Pymble Ladies' College

HIGHER SCHOOL CERTIFICATE

TRIAL EXAMINATION

2016

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Use pencil for Questions 1-10.
- Write using a black or blue pen for Questions 11-16. Black pen is preferred.
- Board approved calculators may be used.
- A reference sheet is provided.
- In Questions 11-16, show relevant mathematical reasoning and/or calculations.

Total Marks – 100

Section I Pages 1-5

10 marks

- Attempt Questions 1-10
- Allow about 15 mins for this section

Section II Pages 7-16

90 marks

- Attempt Questions 11-16
- Allow about 2 hours 45 minutes for this section

Mark	/100
Highest Mark	/100
Rank	

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Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section.

Use the multiple choice answer sheet for Questions 1-10.

- 1 The line with equation $2y = 3x + 5$ is perpendicular to the line with equation $y = kx$. What is the value of k ?

- (A) $-\frac{3}{2}$
- (B) $-\frac{2}{3}$
- (C) $\frac{2}{3}$
- (D) $\frac{3}{2}$

- 2 A function f , defined on a suitable domain, is given by $f(x) = \frac{6x}{x^2 + 6x - 16}$. What restrictions are there on the domain of f ?

- (A) $x \neq -8$ or $x \neq 2$
- (B) $x \neq -4$ or $x \neq 4$
- (C) $x \neq 0$
- (D) $x \neq 10$ or $x \neq 16$

- 3 The functions f and g are defined by $f(x) = x^2 + 1$ and $g(x) = 3x - 4$, on the set of real numbers.

Which expression is equivalent to $g(f(x))$?

- (A) $3x^2 - 1$
- (B) $9x^2 - 15$
- (C) $9x^2 + 17$
- (D) $3x^3 - 4x^2 + 3x - 4$

4 Given that $f(x) = 4\sin 3x$, what is $f'(0)$?

- (A) 0
- (B) 1
- (C) 12
- (D) 36

5 What is $\int x(3x+2)dx$?

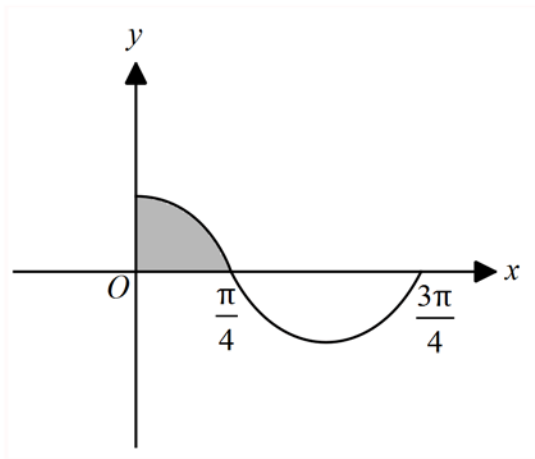
- (A) $x^3 + c$
- (B) $x^3 + x^2 + c$
- (C) $\frac{1}{2}x^2\left(\frac{3}{2}x^2 + 2x\right) + c$
- (D) $3x^2 + 2x + c$

6 If $e^{4t} = 6$, which of the following is an expression for t ?

- (A) $t = \log_e \frac{3}{2}$
- (B) $t = \frac{\log_e 6}{4}$
- (C) $t = \frac{6}{\log_e 4}$
- (D) $t = \frac{\log_e 6}{\log_e 4}$

- 7 The diagram shows part of the graph of $y = a \cos bx$. The shaded area is $\frac{1}{2}$ unit².

What is the value of $\int_0^{\frac{3\pi}{4}} (a \cos bx) \, dx$?

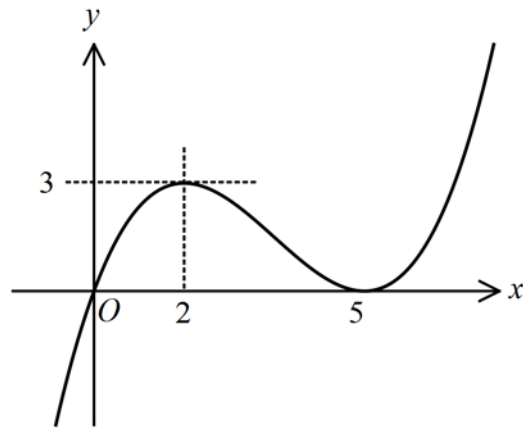


- (A) -1
- (B) $-\frac{1}{2}$
- (C) $\frac{1}{2}$
- (D) $1\frac{1}{2}$

- 8 The volume of a sphere is given by the formula $V = \frac{4}{3}\pi r^3$.
What is the rate of change of V with respect to r , at $r = 2$?

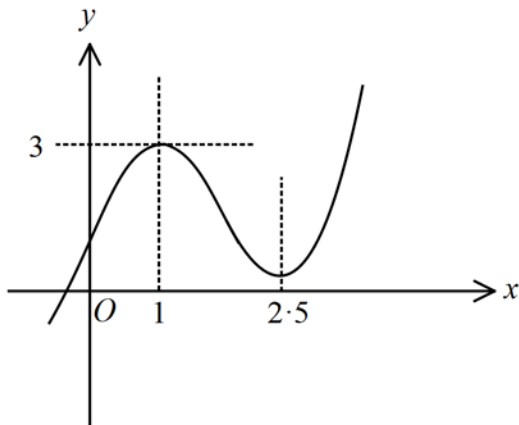
- (A) $\frac{16\pi}{3}$
- (B) $\frac{32\pi}{3}$
- (C) 16π
- (D) 32π

- 9 The diagram shows part of the graph of $y = f(x)$.

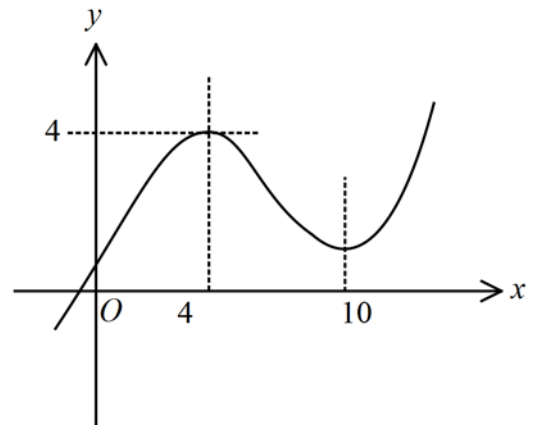


Which of the following diagrams could be the graph of $y = 2f(x) + 1$?

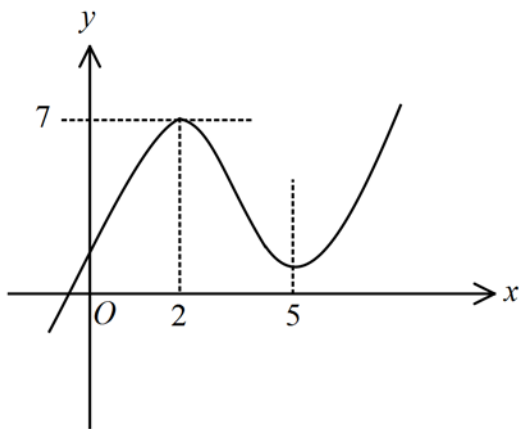
(A)



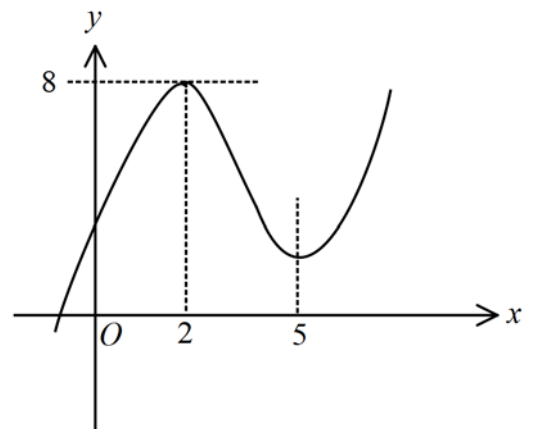
(B)



(C)



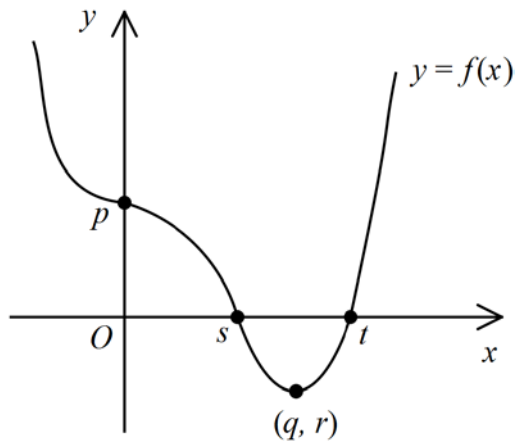
(D)



- 10 The graph of $y = f(x)$ shown has stationary points at $(0, p)$ and (q, r) .

Here are two statements about $f(x)$

- (i) $f(x) < 0$ for $s < x < t$.
- (ii) $f'(x) < 0$ for $x < q$.



Which of the following is true?

- (A) Neither statement is correct.
- (B) Only statement (i) is correct.
- (C) Only statement (ii) is correct.
- (D) Both statements are correct.

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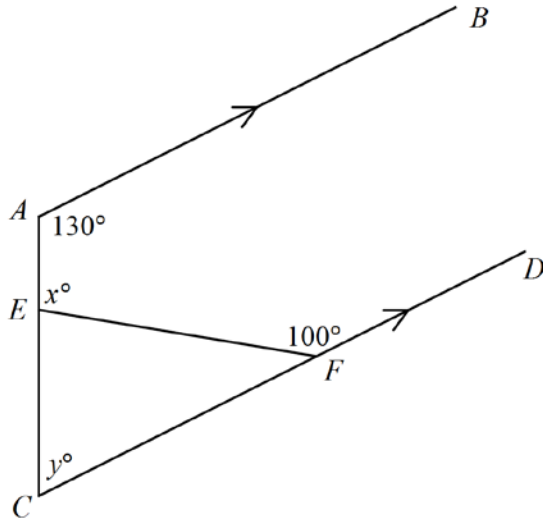
Question 11 (15 marks) Use a separate writing booklet.

- (a) Express $\frac{2}{\sqrt{6}-2}$ with a rational denominator in its simplest form. 2
- (b) Solve $|x-4| \leq 2$. 2
- (c) Solve the equation $\frac{2-x}{3} - \frac{3-x}{2} = \frac{1}{5}$. 2
- (d) Find the gradient of the tangent to the curve $y = (3x+1)^4$ when $x = \frac{1}{3}$. 2
- (e) Simplify $\frac{\log_b a^m}{\log_m a}$ and express it in terms of base b . 2
- (f) Differentiate $\sqrt[3]{x}$. 1
- (g) Find $\int \frac{x+2}{x^2+4x} dx$. 2
- (h) Evaluate $\int_0^{\frac{\pi}{2}} \sec^2\left(\frac{x}{2}\right) dx$. 2

End of Question 11

Question 12 (15 marks) Use a separate writing booklet.

(a)



In the diagram, $AB \parallel CD$, $\angle BAE = 130^\circ$, $\angle EFD = 100^\circ$.

- | | |
|--|----------|
| (i) Find the value of y , giving reasons. | 1 |
| (ii) Find the value of x , giving reasons. | 2 |

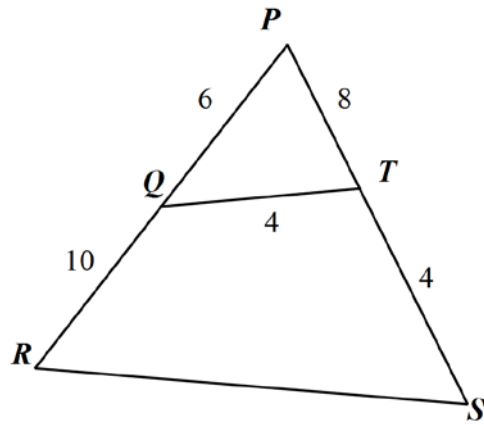
(b) A function f is defined on the set of real numbers by $f(x) = (x - 2)(x^2 + 1)$.

- | | |
|--|----------|
| (i) Find where the graph of $y = f(x)$ cuts | |
| (1) the x -axis. | 1 |
| (2) the y -axis. | 1 |
| (ii) Find the coordinates of the stationary points on the curve with equation $y = f(x)$ and determine their nature. | 4 |
| (iii) Sketch the graph of $y = f(x)$, showing all important features. | 3 |

Question 12 continues on page 9.

Question 12 (continued).

(c)



(i) Prove that $\triangle PQT$ and $\triangle PSR$ are similar.

2

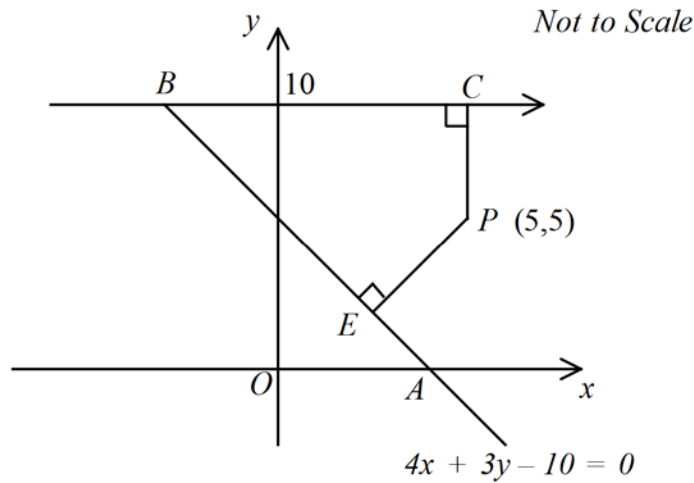
(ii) Hence, find the length of RS .

1

End of Question 12

Question 13 (15 marks) Use a separate writing booklet.

(a)



In the diagram above, the equations of the lines BE and BC are $4x + 3y - 10 = 0$ and $y = 10$ respectively.

P is the point $(5, 5)$.

$PE \perp BE$, and $BC \perp PC$.

- (i) Show that the perpendicular distance from P to BE is 5 units. 1
- (ii) Hence prove that $\triangle BCP \equiv \triangle BEP$. 3
- (iii) Show that the coordinates of B are $(-5, 10)$. 1
- (iv) Show that the locus of points which are equidistant from the lines BC and BE is given by the equation $x + 2y - 15 = 0$. 2

Question 13 continues on page 11

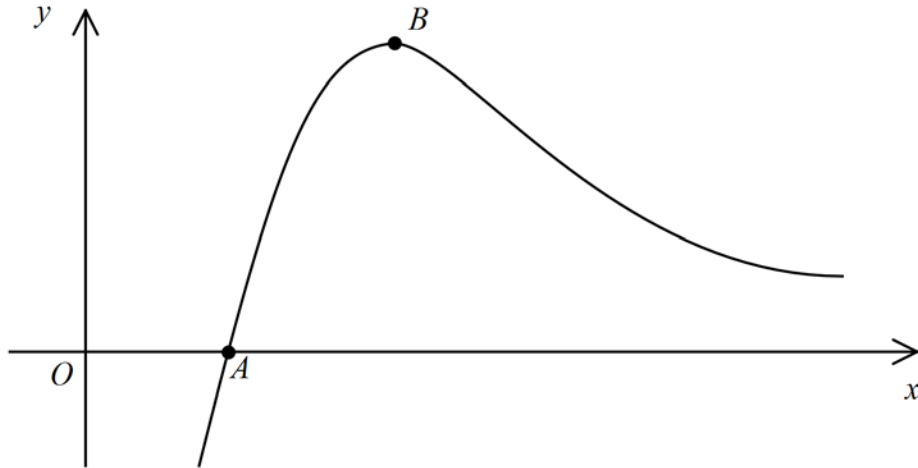
Question 13 (continued).

- (b) (i) Sketch the parabola P , whose focus is $(-2, 2)$ and whose directrix is the line $x = -6$. 2
Indicate on your diagram the coordinates of the focus, the vertex and the equation of the directrix.
- (ii) Determine the equation of the parabola, P . 1
- (c) For what values of a will the equation $ax^2 + 5x + a$ be positive definite? 3
- (d) A ball is dropped from a height of 10 metres and each time it bounces, it reaches $\frac{4}{5}$ of its previous height. What is the total distance travelled by the ball? 2

End of Question 13

Question 14 (15 marks) Use a separate writing booklet.

- (a) The diagram show the curve $y = \frac{\ln x}{x}$.



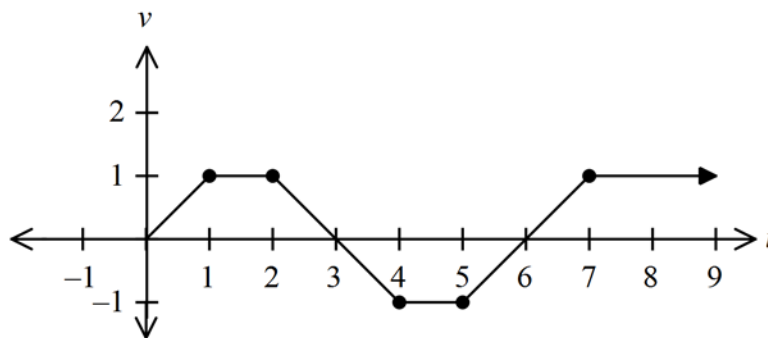
The curve crosses the x -axis at A and has a stationary point at B .

- | | | |
|-------|--|---|
| (i) | State the coordinates of A . | 1 |
| (ii) | Find the coordinates of the stationary point B , of the curve, giving your answer in an exact form | 2 |
| (iii) | Find the exact value of the equation of the normal to the curve at the point where $x = e^3$. | 2 |
-
- | | | |
|-----|---|---|
| (b) | Sketch the graph $y = 2 - \cos 2x$ for $-\pi \leq x \leq \pi$. | 2 |
|-----|---|---|

Question 14 continues on page 13

Question 14 (continued).

(c)



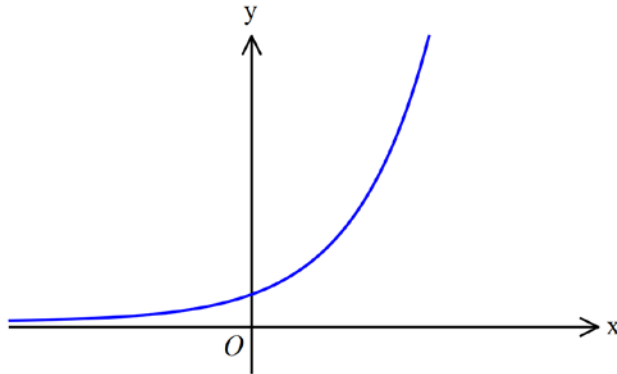
The diagram above describes the velocity, v , of an electrical pulse along a wire in metres/second. Refer to the diagram and answer the questions below.

- | | | |
|-------|--|----------|
| (i) | When is the pulse travelling in a positive direction? | 1 |
| (ii) | When is the pulse stopped? | 1 |
| (iii) | Describe the motion of the particle for $0 < t < 3$. | 2 |
| (iv) | Find the area between the curve and the t axis for $3 \leq t \leq 6$. | 1 |
-
- | | | |
|------|--|----------|
| (d) | If α and β are the roots of the quadratic equation $3x^2 - 4x - 1 = 0$, find | |
| (i) | $\alpha + \beta$ and $\alpha\beta$. | 1 |
| (ii) | $(\alpha + 2)(\beta + 2)$ | 2 |

End of Question 14

Question 15 (15 marks) Use a separate writing booklet.

- (a) The diagram shows a sketch of the curve $y = 2^{4x}$.

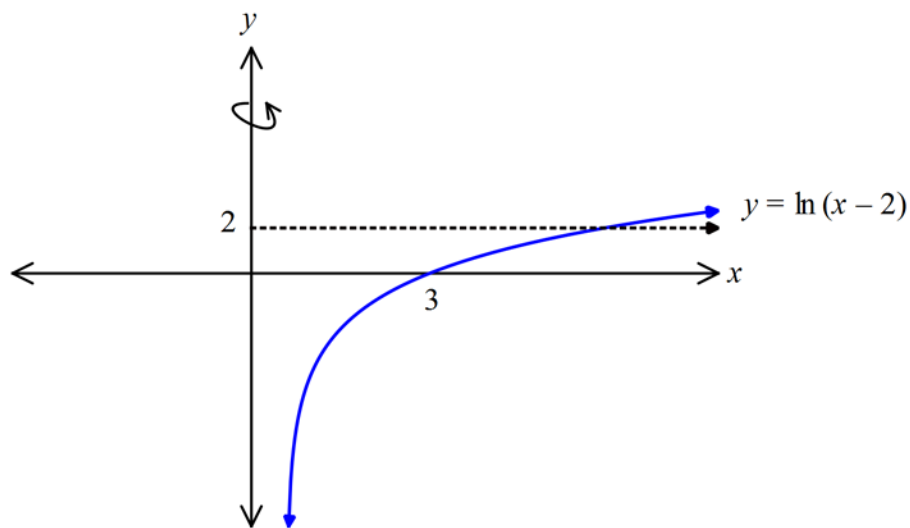


- (i) Use the trapezoidal rule with three function values to find an approximate value for $\int_0^1 2^{4x} dx$. 2
- (ii) Is the approximate for $\int_0^1 2^{4x} dx$, an under approximation or an over approximation? Explain your choice. 1
- (b) The concentration of the pesticide, *Xpesto*, in soil can be modelled by the equation
- $$P_t = P_0 e^{-kt}$$
- where
- P_0 is the initial concentration
 - P_t is the concentration at time t
 - t is the time, in days, after the application of the pesticide.
- (i) Once in the soil, the half-life of a pesticide is the time taken for it's concentration to be reduced to one half of its initial value. If the half-life of *Xpesto* is 25 days, find the value of k correct to 2 significant figures. 2
- (ii) Eighty days after the initial application, what is the percentage decrease in concentration of *Xpesto*? 3

Question 15 continues on page 15

Question 15 (continued).

(c)



The region bounded by the curve $y = \ln(x - 2)$ and the y-axis between $y = 0$ and $y = 2$ is rotated about the y-axis to form a solid.

4

Find the exact volume of the solid.

(d) Given that $\int_0^a 5 \sin 3x \, dx = \frac{10}{3}$, $0 \leq a < \pi$, calculate the value of a .

3

End of Question 15

Question 16 (15 marks) Use a separate writing booklet.

(a) Solve $2 \tan x \sin^2 x = \tan x$ for $0 \leq x \leq 2\pi$. **3**

(b) An open cylindrical can is to have a surface area of $20\pi \text{ cm}^2$. The can has no lid.

(i) Let r centimetres be the radius of the can and h centimetres be its height. **1**

Show that $h = \frac{20 - r^2}{2r}$.

(ii) Hence, show that the total volume of the can is given **1**

by $V = 10\pi r - \frac{1}{2}\pi r^3$.

(iii) Show that the maximum volume is obtained when the height of the can is equal to its radius. **4**

(c) (i) Show that the equation $4\operatorname{cosec}^2\theta - \cot^2\theta = k$, where $k \neq 4$, can be written in the form **2**

$$\sec^2\theta = \frac{k-1}{k-4}.$$

(ii) Hence, or otherwise, solve the equation **4**

$$4\operatorname{cosec}^2(2x + 75^\circ) - \cot^2(2x + 75^\circ) = 5,$$

giving all values of x in the interval $0^\circ < x < 180^\circ$.

End of paper

Year 12 Mathematics Trial solutions

Multiple Choice

- | | |
|------|-------|
| 1. B | 6. B |
| 2. A | 7. B |
| 3. A | 8. C |
| 4. C | 9. C |
| 5. B | 10. B |

Question 11

a) $\frac{2}{\sqrt{b}-2} \times \frac{\sqrt{b}+2}{\sqrt{b}+2}$ ① correct multiplier

$$= \frac{2\sqrt{b}+4}{b-4}$$

$$= \frac{2\sqrt{b}+4}{2}$$

$$= \sqrt{b}+2$$

① correct simplification

b) $|x-4| \leq 2$

$$-2 \leq x-4 \leq 2$$

① correct boundaries

$$2 \leq x \leq 6$$

① correctly expressing as a closed interval

c) $\frac{2-x}{3} - \frac{3-x}{2} = \frac{1}{5}$

$$10(2-x) - 15(3-x) = 6$$

$$20 - 10x - 45 + 15x = 6$$

— ① correctly removing fraction, or equivalent

d) $y = (3x+1)^4$

$$\frac{dy}{dx} = 4(3x+1)^3 \times 3$$

$$= 12(3x+1)^3 \quad \text{— ① correct derivative}$$

When $x = \frac{1}{3}$ $\frac{dy}{dx} = 12(3 \times \frac{1}{3} + 1)^3$

$$= 96 \quad \text{— ① correct substitution and evaluation}$$

e) $\frac{m \log_b a}{\log_m a}$

$$= m \log_b a \div \frac{\log_b a}{\log_b m} \quad \text{— ① correct change of base and multiplying}$$

$$= m \log_b a \times \frac{\log_b m}{\log_b a}$$

$$= m \log_b m$$

$$= \log_b m^m$$

} ① for either answer

f) $y = x^{1/3}$

$$\frac{dy}{dx} = \frac{1}{3} x^{-2/3} \quad \text{— ① r/w}$$

$$= \frac{1}{3\sqrt[3]{x^2}}$$

g) $\int \frac{x+2}{x^2+4x} dx$

① — for $1/2$

$$= \frac{1}{2} \ln |x^2+4x| + C$$

① — correct log and constant

$$\begin{aligned}
 & \int_{\pi/2}^0 \sec^2\left(\frac{x}{2}\right) dx \\
 &= 2 \left[\tan \frac{x}{2} \right]_0^{\pi/2} - 1 \\
 &= 2 \left\{ \tan \frac{\pi}{4} - \tan 0 \right\} - 1 \\
 &= 2 - 1 = 1
 \end{aligned}$$

Q

Question 12

(a)(i) $y = 180^\circ - 130^\circ = 50^\circ$ (co-interior angles supplementary AB || CD)

(1)

(ii) $\angle EFC = 180^\circ - 100^\circ = 80^\circ$ (adjacent supplementary angles)

$x^\circ = y^\circ + \angle EFC$ (exterior angle of $\triangle EFC$ equals sum of 2 opposite interior angles).
 $= 50^\circ + 80^\circ = 130^\circ$

(2)

* some found $\angle CEF$ Assume when angles on straight line

(b)(i) $f(x) = (x-2)(x^2+1)$

(1) cuts x axis $y=0$
 x^2+1 no solution
 $(2, 0)$

(1)

(2) cuts y axis $x=0$
 $y = (0-2)(0^2+1) = -2$
 $(0, -2)$

(1)

(ii) let $f(x) = y$
 $y = (x-2)(x^2+1)$
 $= x^3 - 2x^2 + x - 2$
 $y' = 3x^2 - 4x + 1$
 stat pts can occur $y' = 0$
 $0 = 3x^2 - 4x + 1$
 $0 = (x-1)(3x-1)$

$$x=1 \quad y = (1-2)(1^2+1) \\ = -2 \quad (1, -2)$$

$$x = \frac{1}{3} \quad y = \left(\frac{1}{3}-2\right)\left(\left(\frac{1}{3}\right)^2+1\right) \\ = -\frac{50}{27} \quad \left(\frac{1}{3}, -\frac{50}{27}\right)$$

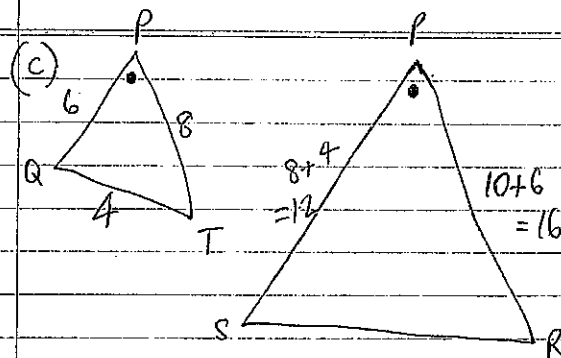
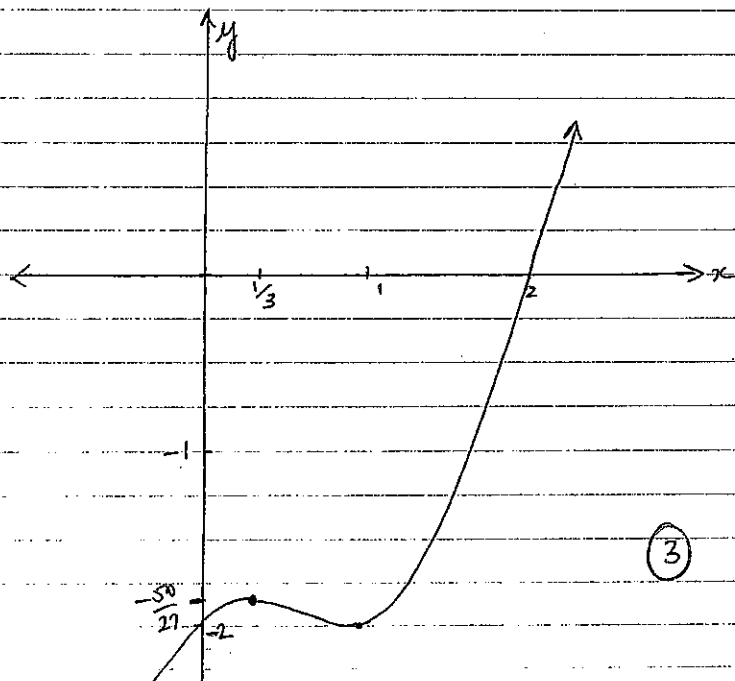
use y'' to determine nature
* if you use y' as test
you must show substitutions
and values.

$$y'' = 6x - 4$$

$$x=1 \quad y'' = 6(1) - 4 \\ = 2 > 0 \quad \therefore (1, -2) \text{ minimum.} \quad (4)$$

$$x = \frac{1}{3} \quad y'' = 6\left(\frac{1}{3}\right) - 4 \\ = -2 < 0 \quad \therefore \left(\frac{1}{3}, -\frac{50}{27}\right) \text{ maximum.}$$

(iii)



(i) In $\triangle PQT$ and $\triangle PSR$

$$\angle QPT = \angle SPR \quad (\text{common angle})$$

$$\frac{PQ}{PS} = \frac{6}{12} = \frac{1}{2} \quad \text{given} \quad (1) \text{ ratios}$$

$$\frac{PT}{PR} = \frac{8}{16} = \frac{1}{2} \quad \text{given} \quad (1) \text{ reason}$$

$\therefore \triangle PQT \parallel \triangle PSR$ (2 pairs of matching sides
in proportion and included
angle equal)

$$(ii) \quad \frac{QT}{SR} = \frac{1}{2} = \frac{4}{8}$$

$\therefore x = 8$ (1) matching sides in similar $\triangle$ s.

Question 13

a) $4x + 3y - 10 = 0$
Point $(5, 5)$

perp dist = $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

= $\frac{|4 \times 5 + 3 \times 5 - 10|}{\sqrt{4^2 + 3^2}}$

= $\frac{|20 + 15 - 10|}{\sqrt{25}}$

= $\frac{25}{5}$

= 5 units

①

iii. C is $(5, 10)$, 5 units vertically above P $(5, 5)$.

Construct BP.

① $CP = EP$

In $\triangle BCP$ and $\triangle BEP$

$\angle BCP = \angle BEP$ (given)

BP is common

$CP = EP$ (part (i) and above, both 5 units)

$\therefore \triangle BCP \equiv \triangle BEP$ (RHS)

iii. B lies on the line $y = 10$,
 $\therefore y$ coordinate of B is 10.

$4x + 3y - 10 = 0$

when $y = 10$

$4x + 3(10) - 10 = 0$

$4x + 20 = 0$

$4x = -20$

$x = -5$

①

iv. As established in (ii), P is equidistant from line BA and line BC.
 $\therefore$ The locus passes through B and P.

$m_{BP} = \frac{y_2 - y_1}{x_2 - x_1}$

= $\frac{5 - 10}{5 - 5}$

= $\frac{5 - (-5)}{5 - 10}$

= $\frac{-10}{-5}$

= 2

$y - y_1 = m(x - x_1)$

$y - 5 = 2(x - 5)$

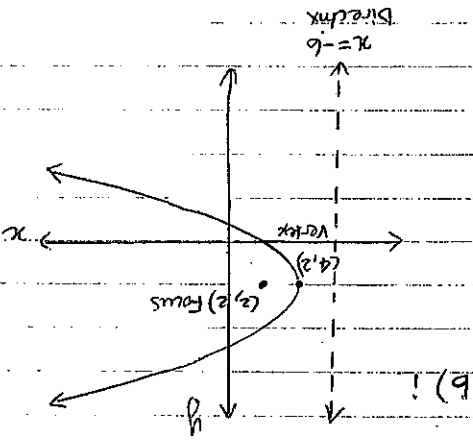
$2y - 10 = -x + 5$

$x + 2y - 15 = 0$

(several methods possible)

① progress

① correct locus



b)!

② all correct
-1 each error

①

ii. Focal length = 2

$(y - 2)^2 = 4a(x + 4)$

$(y - 2)^2 = 8(x + 4)$

c) For $ax^2 + 5x + a$ to be positive definite,

$$a > 0, \Delta < 0.$$

$$\Delta = b^2 - 4ac$$

$$= 5^2 - 4 \times a \times a$$

$$= 25 - 4a^2$$

$$25 - 4a^2 < 0$$

①

$$25 < 4a^2$$

$$4a^2 > 25$$

$$a^2 > \frac{25}{4}$$

4

$$a < -\frac{5}{2}, a > \frac{5}{2}$$

①

2 2

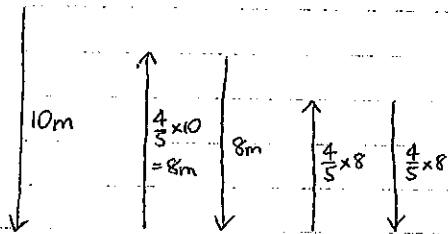
Since $a > 0$,

$a > \frac{5}{2}$ is the only soln.

①

2

d)



$$a = 8$$

$$r = \frac{4}{5}$$

$$(|r| < 1)$$

etc.

$$\text{Total distance} = 10 + 2 \times S_{\infty}$$

$$= 10 + 2 \times \frac{a}{1-r}$$

$$= 10 + 2 \times \frac{8}{1 - \frac{4}{5}}$$

$$= 90\text{m}$$

① S_{∞}

① Total distance

Question 14

$$a) \quad y = \frac{\ln x}{x}$$

(i) y-coordinate of A is 0

$$\therefore 0 = \frac{\ln x}{x}$$

$$x$$

$$\ln x = 0$$

$$x = e^0$$

$$x = 1$$

$\therefore A(1, 0)$ (i) R/W

$$(ii) \quad y = \frac{\ln x}{x} \quad u = \ln x \quad v = x$$

$$\frac{dy}{dx} = \frac{v u' - u v'}{v^2}$$

$$= \frac{x \times \frac{1}{x} - \ln x \times 1}{x^2}$$

$$= \frac{1 - \ln x}{x^2} \quad (1)$$

Find x when $\frac{dy}{dx} = 0$ for stationary point.

$$0 = \frac{1 - \ln x}{x^2}$$

$$1 - \ln x = 0$$

$$\ln x = 1$$

$$x = e^1$$

$$x = e$$

$$y = \frac{\ln e}{e}$$

$$= \frac{1}{e}$$

$\therefore B(e, \frac{1}{e})$ (i)

(iii)

when $x = e^3$

$$\frac{dy}{dx} = \frac{1 - \ln e^3}{(e^3)^2}$$

$$= \frac{1 - 3}{e^6}$$

$$M_T = -2e^{-6}$$

This is gradient of tangent
so gradient of normal is

$$M_N = -\frac{1}{M_T}$$

$$M_T$$

$$= -\frac{1}{-2e^{-6}}$$

$$= \frac{2e^6}{1}$$

$$= \frac{e^6}{2} \quad (1)$$

$$y = \frac{\ln e^3}{e^3}$$

$$= 3e^{-3}$$

$\therefore$ equation of normal is
 $y - 3e^{-3} = \frac{e^6}{2}(x - e^3)$ (1)

$$2y - 6e^{-3} = e^6 x - e^9$$

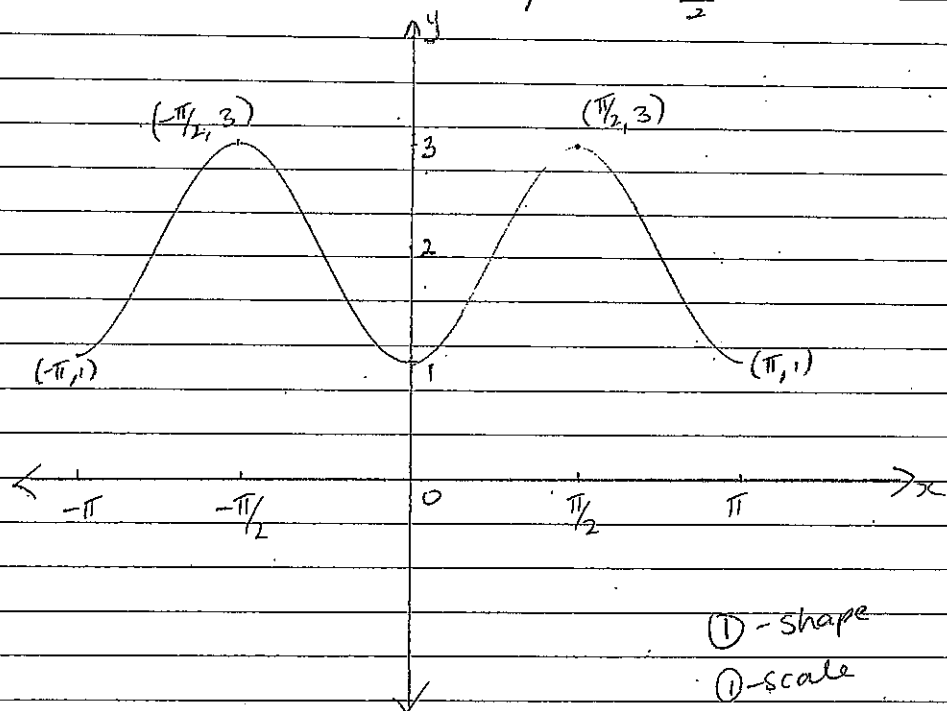
$$e^6 x - 2y + 6e^{-3} - e^9 = 0$$

$(x e^3)$

$$e^9 x - 2e^3 y + 6 - e^{12} = 0$$

$$y = \frac{e^9 x - \frac{6}{e^3} - \frac{2}{e^9}}{\frac{2}{e^3}} + \frac{e^3}{2}$$

b) $y = 2 - \cos 2x$ $-\pi \leq x \leq \pi$
 period = $\frac{2\pi}{2} = \pi$



c) (i) $0 < t < 3$ and $t > 6$ ①

(ii) $t = 0, 3, 6$ ①

(iii) Particle moves in positive direction for $0 < t < 3$

During $0 < t < 1$ particle has constant acceleration so speed increases.

During $1 \leq t \leq 2$ particle has zero acceleration so speed is constant at 1 m/s

During $2 < t < 3$ particle has constant deceleration so speed decreases

① - describe direction

① - describe speed.

(iv) $A = \frac{1}{2}h(a+b)$
 $= \frac{1}{2} \times 1 \times (1+3)$
 $= 2 \text{ units}^2$ ①

d) $3x^2 - 4x - 1 = 0$

$a = 3$ $b = -4$ $c = -1$

(i) $\alpha + \beta = -\frac{b}{a}$ $\alpha\beta = \frac{c}{a}$
 $= -\frac{(-4)}{3}$ $= -\frac{1}{3}$
 $= \frac{4}{3}$ ①

(ii) $(\alpha+2)(\beta+2)$
 $= \alpha\beta + 2(\alpha+\beta) + 4$ ①
 $= -\frac{1}{3} + 2 \times \frac{4}{3} + 4$
 $= \frac{19}{3}$ ①

2 unit 1K19L

do not
use 2 column
or arrows in
your solutions.

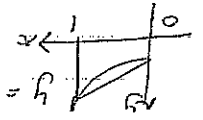
15	a	x	0	1/2	1
		y	1	4	16
		Using	$a = \frac{2}{3}(a+b)$		

Area = $\frac{2}{3} \left(1 + 4 + \frac{16}{4} \right) = \frac{2}{3} \times 25 = \frac{50}{3}$

$\frac{50}{3} (4 + 16)$

2

(ii) Area under the curve is greater than
area under concave up curves as shown
in diagram



(b)(i) $\frac{d}{dt} P_0 = P_0 e^{-kt}$

$\frac{1}{t} = e^{-k(25)} \quad (t = 25)$

$\ln\left(\frac{1}{t}\right) = -25k$

$k = -\frac{1}{25} \ln\left(\frac{1}{t}\right)$

$= 0.027725 \dots$

$= 0.028$ (2 sig. figures) ✓

(ii) $t = 80$ $P = P_0 e^{-k(80)}$

$0.027725 \dots$ should use

10.88% remaining for k

$89.1181177 \dots$ used

89% (correct to nearest whole no.)

(c) $y = \ln(x-2)$

$e^y = x-2$

$x = 2 + e^y$

$x^2 = (2 + e^y)^2$

$= 4 + 4e^y + e^{2y}$

* Very important stuff

$\cos 3a = 1$
 $\cos 3a = -1$
 $3a = \pi$
 $a = \frac{\pi}{3}$

$0 \leq 3a < 3\pi$

$-\frac{3}{5} [\cos 3a - \cos 0] = \frac{3}{10}$

$-\frac{3}{5} \cos 3a = \frac{3}{10}$

(d) $\int_0^a 5 \sin 3x \, dx = \frac{3}{10}$

$\frac{5}{3} [e^4 + 8e^2 + 7]$

4

$\pi [3\frac{1}{2} + 4e^2 + \frac{7}{e^4}]$

$\pi [(8 + 4e^2 + \frac{7}{e^4}) - (0 + 4 + \frac{7}{e^4})]$

$\pi [(4(2) + 4e^2 + \frac{7}{e^4}) - (4(0) + 4e^2 + \frac{7}{e^4})]$

$\pi [4y + 4e^y + \frac{7}{e^{2y}}]_2^0$

$\pi \int_2^0 (4 + 4e^y + e^{-2y}) \, dy$

(c) $V = \pi \int_2^0 (2 + e^y)^2 \, dy$

Question 16

a) $2 \tan x \sin^2 x - \tan x = 0$

$$\tan x (2 \sin^2 x - 1) = 0$$

$\therefore \tan x = 0$ or $\sin^2 x = \frac{1}{2}$ ① correct separation

$$x = 0, \pi, 2\pi \text{ or } \sin x = \pm \frac{1}{\sqrt{2}}$$

① correct values $x = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4$

for $\tan x = 0$ ① correct values for $\sin x = \pm \frac{1}{\sqrt{2}}$

b) i) $20\pi = 2\pi r h + \pi r^2$

$$1. 20 = 2rh + r^2$$

$$2rh = 20 - r^2$$

$$h = \frac{20 - r^2}{2r} \quad \text{① - correct demonstration}$$

ii) $V = \pi r^2 h$
 $= \pi r^2 \left(\frac{20 - r^2}{2r} \right)$

$$= \frac{\pi r (20 - r^2)}{2}$$

$$= \frac{20\pi r}{2} - \frac{\pi r^3}{2} \quad \text{① - correct demonstration}$$

$$V = 10\pi r - \frac{1}{2}\pi r^3$$

ii) $\frac{dV}{dr} = 10\pi - \frac{3}{2}\pi r^2$

$$\frac{d^2V}{dr^2} = -3\pi r$$

when $\frac{dV}{dr} = 0$ ① differentiating and making

$$\frac{3}{2}\pi r^2 = 10\pi$$

$$r^2 = \frac{20}{3}$$

$$r = \sqrt{\frac{20}{3}} \quad \text{--- ① Solving for } r$$

When $r = \sqrt{\frac{20}{3}} \quad \frac{d^2V}{dr^2} = -3\pi \times \sqrt{\frac{20}{3}} < 0$

$\therefore r = \sqrt{\frac{20}{3}}$ will produce maximum volume.

$$r = \frac{\sqrt{20}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\ = \frac{2\sqrt{5} \times \sqrt{3}}{3} \\ = \frac{2\sqrt{15}}{3}$$

From i)

$$h = \frac{20 - \left(\frac{2\sqrt{15}}{3}\right)^2}{2\left(\frac{2\sqrt{15}}{3}\right)}$$

$$= \frac{20 - \frac{60}{9}}{\frac{4\sqrt{15}}{3}}$$

$$= \frac{120}{9} \times \frac{3}{4\sqrt{15}}$$

$$= \frac{10}{\sqrt{15}} \times \frac{\sqrt{15}}{\sqrt{15}}$$

$$= \frac{10\sqrt{15}}{15}$$

$$= \frac{2\sqrt{15}}{3}$$

$$= r$$

$\therefore$ Volume is maximum when $h = r$

c) i) $4 \operatorname{cosec}^2 \theta - \cot^2 \theta = k$

$$\frac{4}{\sin^2 \theta} - \frac{\cos^2 \theta}{\sin^2 \theta} = k$$

$$4 - \cos^2 \theta = k \sin^2 \theta \quad \text{--- ① to this point or equivalent}$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$4 - k = \cos^2 \theta - k \cos \theta$$

$$(k - 1) \cos^2 \theta = k - 4$$

$$\cos^2 \theta = \frac{k - 4}{k - 1}$$

$$\frac{1}{1} = \frac{-(k - 4)}{-(k - 1)}$$

$$\sec^2 \theta = \frac{k - 1}{k - 4}$$

① suitable demonstration

$$\sec^2 (2x + 75) = \frac{5 - 1}{5 - 4}$$

$$\therefore \sec^2 (2x + 75) = 4 \quad \text{--- ① linking to 1)$$

$$0^\circ < x < 180^\circ$$

$$75^\circ < 2x + 75 < 435^\circ \quad \text{--- ① Adjustment of domain}$$

$$\therefore \frac{1}{4} = \frac{\cos^2 (2x + 75)}{4}$$

$$\cos^2 (2x + 75) = \frac{4}{4}$$

$$\cos (2x + 75) = \pm \frac{1}{2} \quad \text{--- ① Correctly square rooting and changing to cos or equivalent}$$

$$2x + 75 = 120^\circ, 240^\circ, 300^\circ, 420^\circ$$

$$x = 22.5^\circ, 82.5^\circ, 112.5^\circ, 172.5^\circ \quad \text{--- ① correct solution}$$